A/B Test Analysis for an E-Commerce Website: Comparing Two Pricing Strategies Using Hypothesis Testing

Mini Project Report

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Abstract

This project evaluates whether a new pricing/landing page (the "treatment") converts website visitors better than the existing page (the "control"), using a real e-commerce A/B test dataset of **294,478** user sessions. After cleaning the data (removing mismatched group/page records and duplicate users), **290,584** sessions across three countries (United States, United Kingdom, Canada) were analyzed. Three complementary statistical tests were applied: an independent-samples **t-test**, a **chi-square test of independence**, and **one-way ANOVA**. All three tests agree: there is **no statistically significant difference** in conversion rate between the two pricing strategies ($p \approx 0.19$), and this holds even when the data is broken down by country. Based on this evidence, the recommendation is to **retain the existing pricing page** rather than incur the cost and risk of switching to a new page that shows no measurable benefit.

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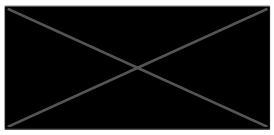
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1 Introduction

E-commerce companies routinely test changes to their websites before rolling them out to every customer. This project analyzes a controlled experiment in which visitors to an e-commerce site were randomly assigned to one of two groups:

- **Control group** – shown the existing ("old") pricing page.
- **Treatment group** – shown a new pricing page.

For every visit, the dataset records whether the user *converted* (made a purchase, coded as 1) or not (coded as 0). The business question this report answers is:

> *Does the new pricing strategy produce a higher conversion rate than the old one, with statistical confidence – or is any observed difference simply due to chance?*

To answer this rigorously, three classical hypothesis-testing techniques are applied and cross-checked against one another:

1. An **independent two-sample t-test** comparing mean conversion between groups.
2. A **chi-square test of independence** on the group × conversion contingency table.
3. A **one-way ANOVA** testing whether conversion differs across countries, and across the six country/strategy combinations.

2 Data Description

Two files were provided:

- `ab_data.csv` – 294,478 rows, with columns `user_id`, `timestamp`, `group` (*control*/*treatment*), `landing_page` (*old_page*/*new_page*), and `converted` (0/1).
- `countries.csv` – 290,584 rows mapping `user_id` to a `country` (US, UK, or CA).

2.1 Data Cleaning

Two data-quality issues were identified and corrected before analysis:

1. **Group/page mismatches.** In a clean experiment, every *control* user should see the *old_page* and every *treatment* user should see the *new_page*. **3,893** rows violated this rule (e.g., a control-group user who was shown the new page) and were removed, since it is impossible to know which group these users truly belong to.
2. **Duplicate users. 3,894** `user_id` values appeared more than once. Only the first recorded visit for each user was retained, so that no single user is counted twice.

After cleaning and merging with the country data, **290,584** sessions remained, split almost evenly between the two groups (Figure 1).

Table 1: Sample sizes after cleaning

Group	Users (n)	Conversions
Control (old page)	145,274	17,489
Treatment (new page)	145,310	17,264
Total	**290,584**	**34,753**

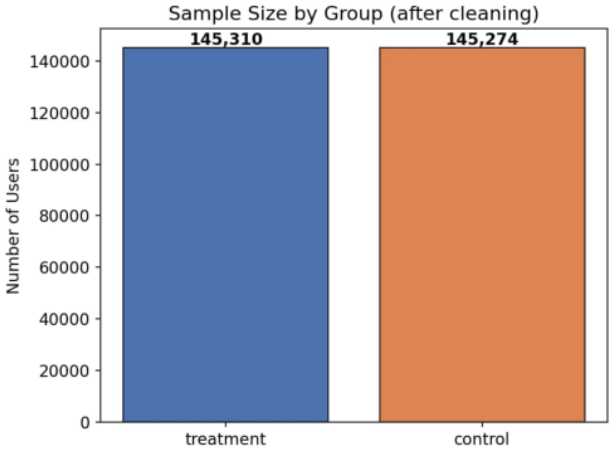


Figure 1: Sample size per group after cleaning. The near-identical group sizes confirm the random assignment worked as intended.

3 Methodology

All tests use a significance threshold of $\alpha = 0.05$. For each test, the null hypothesis (H_0) represents "no effect / no relationship," and the alternative hypothesis (H_1) represents "there is an effect." A p-value below 0.05 leads us to reject H_0 in favor of H_1.

3.1 Test 1: Independent t-test

Since `converted` is a 0/1 variable, its mean is exactly the conversion rate. Welch's independent two-sample t-test (which does not assume equal variances) was used to compare the mean conversion rate of the treatment group against the control group.

$$H_0 : \mu_{\text{treatment}} = \mu_{\text{control}} \qquad H_1 : \mu_{\text{treatment}} \neq \mu_{\text{control}}$$

3.2 Test 2: Chi-square test of independence

Because both `group` and `converted` are categorical, a chi-square test on the 2×2 contingency table (group vs. converted) provides a natural, distribution-free check on whether conversion depends on which page a visitor saw.

H_0 : strategy and conversion are independent $\qquad$ H_1 : strategy and conversion are related

3.3 Test 3: One-way ANOVA

To check whether geography plays a role, a one-way ANOVA compared mean conversion rates across the three countries (US, UK, CA), and separately across the six country $\times$ strategy combinations (e.g., "control – US", "treatment – UK", etc.).

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k \qquad H_1 : \text{at least one group mean differs}$$

4 Results

4.1 Descriptive Overview

Figure 2 shows the observed conversion rate in each group. The control group converted at **12.04%** and the treatment group at **11.88%** – a raw difference of just **−0.16 percentage points** in favor of the old page.

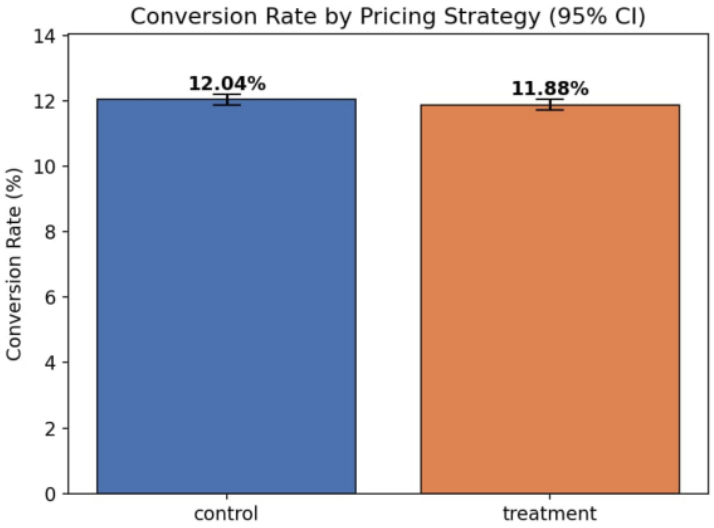


Figure 2: Conversion rate by pricing strategy, with 95% confidence intervals. The intervals overlap substantially, hinting the difference may not be statistically meaningful.

4.2 T-Test Results

Table 2: Welch's t-test: control vs. treatment conversion

Metric	**Value**
Control mean conversion	12.039%
Treatment mean conversion	11.881%
Difference (treatment − control)	−0.158 pp
95% CI of difference	[−0.394%, 0.078%]
t-statistic	−1.311
p-value	**0.190**

Since $p = 0.190 > 0.05$, we **fail to reject** H_0. The 95% confidence interval for the difference in conversion rates spans zero, confirming there is no statistically significant evidence that the new page changes conversion in either direction. The effect size (Cohen's $d \approx -0.005$) is negligible even before considering significance.

4.3 Chi-Square Test Results

Table 3: Contingency table: strategy vs. conversion outcome

Group	Not Converted	Converted
Control	127,785	17,489
Treatment	128,046	17,264

Table 4: Chi-square test of independence

Metric	Value
χ^2 statistic	1.704
Degrees of freedom	1
p-value	**0.192**

The chi-square test agrees with the t-test: $p = 0.192 > 0.05$, so we **fail to reject** H_0. The observed counts are very close to what would be expected if conversion were completely independent of which page a visitor saw.

4.4 ANOVA Results

Table 5: Conversion rate by country

Country	Users (n)	Conversion Rate
Canada (CA)	14,499	11.53%
United Kingdom (UK)	72,466	12.06%
United States (US)	203,619	11.95%

Table 6: One-way ANOVA results

Comparison	F-statistic	p-value
Conversion across 3 countries	1.605	0.201
Conversion across 6 country × strategy groups	1.466	0.197

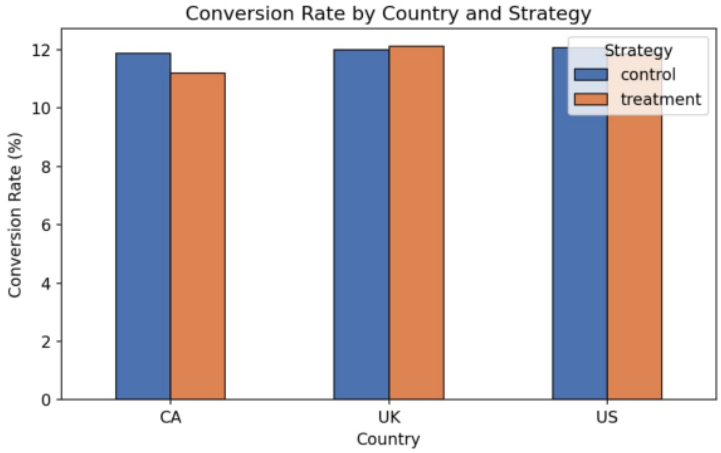


Figure 3: Conversion rate by country and strategy. No country shows a consistent advantage for either page.

Both ANOVA results give $p > 0.05$, so we **fail to reject** H_0 in each case. Conversion rate does not differ significantly across countries, and no specific country/strategy combination stands out as a hidden winner.

5 Discussion

All three tests despite examining the data from different statistical angles reach the same conclusion, which is a good sign of robustness rather than a coincidence:

- The **t-test** shows no significant difference in mean conversion between the two pricing pages.
- The **chi-square test** shows conversion is statistically independent of which page a visitor saw.
- The **ANOVA** shows the picture doesn't change when the population is split by country there is no hidden sub-population where the new page clearly wins.

It is important to note what "not statistically significant" does *not* mean: it does not prove the two pages perform identically, only that this dataset does not provide sufficient evidence of a difference. With roughly 290,000 users and a baseline conversion rate near 12%, this experiment has reasonable power to detect a moderate effect, but a very small true effect (e.g., a 0.1–0.2 percentage-point lift) could plausibly go undetected.

6 Conclusion and Recommendation

Based on rigorous statistical testing across three independent methods, there is **no evidence that the new pricing page improves conversion** over the existing page, either overall or within any individual country.

Recommendation: The company should **retain the current (old) pricing page**. Switching to the new page would introduce engineering and design costs, and risk disrupting returning customers, without any demonstrated benefit to conversion. If the business still believes a small improvement exists, the next step should be a longer-running or larger-sample follow-up test specifically powered to detect a smaller effect size, or testing a more substantially redesigned page rather than a close variant of the current one.

All analysis was performed in Python using `pandas` *for data handling and* `scipy.stats` *for hypothesis testing. The accompanying Jupyter notebook (*`AB_Test_Analysis.ipynb`*) contains the full, reproducible code.*